VITAL: A Verified, Frame-Explicit Flight Simulation Framework that Turns Aircraft Models into Traceable Flying-Qualities Verdicts

Draft abstract and paper outline

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October 29, 2026

Draft abstract (about 250 words)

Flying-qualities problems are usually discovered late, after a configuration is frozen, because the tools that model an aircraft, the tools that simulate it, and the documents that judge it are disconnected. Open flight-dynamics codes simulate well but do not issue verdicts against handling-qualities requirements; conceptual-design environments predict stability but are rarely verified against independent reference data; and clearance methods quantify robustness but are tied to one aircraft and one control law. Certification by analysis needs all three at once: a credible model, a verified simulation, and a rule-based verdict whose margin and uncertainty are stated.

This work presents VITAL, an aircraft-agnostic framework that links these steps: concept, model, trim and linearization, rule verdict, and design feedback. Aircraft are defined as data:

- an explicit tree of reference axes;
- load-producing components;
- mass properties, read from standard DAVE-ML (ANSI/AIAA S-119) files.

Every frame transfer and moment-reference change is therefore performed by one verified mechanism.

Each capability is accepted only through gated, test-first evidence:

- published NASA check cases;
- independent toolbox results;
- analytic solutions;
- deliberate code sabotages that the tests must detect.

The framework is demonstrated on the NASA NESC F-16 model. Early results:

- the published trim is reproduced to within 0.0004°;
- a 10-second Aerospace Blockset 6-DOF hold stays at trim to 10^{-14}, and agrees with the framework's own equations of motion to 10^{-12};
- MIL-F-8785C Class IV levels are assessed with stated margins.

The paper then shows how uncertainty-aware verdicts and sensitivity-based control-law changes restore compliance. It outlines the extension to distributed electric propulsion (NASA X-57) and to FAA Part 23 criteria. [update numbers at submission]

1 Main gap in the literature

- **Simulation and judgement are separate.** Mature simulators (JSBSim [1], MATLAB Aerospace Blockset) integrate the equations of motion but stop at time histories. Turning those into a *verdict* against MIL-F-8785C or Part 23 criteria is left to project-specific scripts.
- **Design-stage stability tools are rarely verified against independent data.**
  - CEASIOM/SDSA [3] and total-envelope conceptual S&C methods [4] predict modes and levels early.
  - SUAVE [2] supports unconventional configurations.
  - None of them reports reproduction of public multi-simulation check cases such as NASA/TM-2015-218675 [6].
- **Robust clearance is aircraft- and law-specific.** GARTEUR clearance methods [5] quantify worst cases, but they are not embedded in a reusable model-to-verdict pipeline. Their "pass" semantics under uncertainty are not standardized.
- **Credibility is the bottleneck for certification by analysis.** NASA's CbA guide [7] and recent credibility studies [8] identify verified models, quantified uncertainty, and traceable evidence as the missing pieces.
- **Frame and reference-point errors are silent.** Moment-reference shifts, canted thrust lines, station/body conventions and inertia products are typically hand-coded per aircraft. For distributed-propulsion aircraft with many canted propulsors, this does not scale and is not auditable.

Gap statement (one sentence). No openly documented framework carries an arbitrary aircraft definition through a simulation that has been verified against independent public reference data, all the way to a rule-based flying-qualities verdict. Such a verdict would state its margin and uncertainty, and would identify the design change that restores compliance.

2 Research question

Main question. Can an aircraft-agnostic simulation framework, verified against independent public reference data, produce *credible and traceable* flying-qualities verdicts, with explicit margins and uncertainty, early enough in design to guide configuration and control-law changes?

Sub-questions.

RQ1. **Representation.** Can an explicit tree of axes, components and mass properties represent conventional and distributed-propulsion aircraft with no aircraft-specific plant code? Can it do so without loss of accuracy or speed against a hand-written model?

RQ2. **Verification.** What evidence is sufficient for each capability to count as credible? The candidates are public check cases, independent tools, analytic solutions and mutation (sabotage) testing.

Table 1: Capabilities of representative tools as described in their published sources (✓ addressed, ○ partial or via add-on, – not addressed). [confirm each cell during the literature review]

Tool / approach	Nonlinear 6-DOF sim	Public check-case verif.	Explicit frames & ref. points	Trim / lin. / modes	Rule-based FQ verdict	Uncert.-aware verdict	Design feed-back	Any config. (e.g. DEP)
JSBSim [1]	✓	○	○	○	–	–	–	✓
Aerospace Blockset	✓	–	○	○	–	–	–	○
SUAVE [2]	–	–	○	○	–	–	✓	✓
CEASIOM / SDSA [3]	✓	–	○	✓	✓	–	○	○
Total-envelope S&C [4]	–	–	–	✓	○	–	✓	○
GARTEUR clearance [5]	✓	–	–	✓	✓	✓	–	–
NESC check cases [6]	✓	✓	–	–	–	–	–	–
VITAL (this work)	✓	✓	✓	✓	✓	✓	✓	○→✓

RQ3. **Verdict.** How should a "pass" be defined when model parameters are uncertain? Specifically, how should worst-case margin and probability of meeting a level be combined so that a robust pass is never weaker than a nominal pass?

RQ4. **Feedback.** Can sensitivities of the verdict margin identify the design or control-law change that restores compliance, confirmed by re-assessment?

3 Proposed paper structure

1. **Introduction.**
   - Why flying-qualities issues surface late.
   - The push toward certification by analysis.
   - New configurations (distributed electric propulsion, eVTOL).
   - Contributions, and the research question.
2. **Related work.**
   - 2.1. *Flight-dynamics simulators*: JSBSim, Aerospace Blockset; what they verify and what they leave to the user.
   - 2.2. *Conceptual-design stability and control*: CEASIOM/SDSA, SUAVE, total-envelope methods.
   - 2.3. *Flight-control clearance and robustness*: GARTEUR AG11, μ-analysis, optimization-based clearance.
   - 2.4. *Simulation credibility and certification by analysis*: NASA CbA guide, NASA-STD-7009 [cite], UQ-based credibility.
   - 2.5. *Gap*: Table 1 and the gap statement.

3. **The VITAL framework.**

3.1. *Frame-explicit object core.*

- Axis tree, points, vectors, loads (wrenches), inertia.
- One verified mechanism for every frame and moment-reference transfer.

3.2. *Standards-based data layer.*

- DAVE-ML import with units and provenance.
- Hash-protected source data.
- Embedded check data reproduced.

3.3. *Aircraft as data.*

- Components (aero database, propulsors) on axes, plus mass properties.
- A compiled fast path proven identical to the object path.

3.4. *Analyses.*

- Typed trim with status codes and stall/data-envelope bounds.
- Linearization with step-plateau check.
- Mode identification.

3.5. *Rule engine.* Cited, machine-readable criteria records (MIL-F-8785C now, FAA Part 23 later); levels and normalized margins over the flight envelope.

3.6. *Uncertainty and verdict logic.*

- Bound-worst margin over a joint-coverage box.
- Monte Carlo at the critical condition.
- A "robust" verdict defined so that it is never weaker than "pass".

3.7. *Simulink and real time.*

- Generated 6-DOF models driven by the same aircraft object.
- Visualization and pilot-in-the-loop.

4. **Verification methodology.**

- Evidence classes: published, independent, analytic, regression.
- Red-green gated development.
- A failure catalogue mapped to tests.
- Sabotage (mutation) testing: every gate must detect deliberate code faults.
- NESC check cases as the external reference.

5. **Case study: NASA F-16.**

5.1. Data provenance: NASA TP-1538 → Stevens & Lewis → NESC DAVE-ML.

5.2. Trim reproduction and the 6-DOF trim hold in Simulink.

5.3. NESC dynamic check cases within the multi-simulation band.

5.4. MIL-F-8785C Class IV levels, critical conditions, pre-registered degradations that must fail.

5.5. Design feedback: a pitch SAS and yaw damper restore the target level, confirmed by re-assessment.

5.6. Uncertainty: robust verdicts and their sensitivity to assumed widths.

6. **Toward distributed electric propulsion (planned).**
   - A twin-engine and engine-out known-answer aircraft.
   - NASA X-57 Maxwell: canted propulsors, gyroscopic and propeller-wing coupling.
   - Part 23 criteria.

7. **Discussion and limitations.**
   - Judgement-based uncertainty widths.
   - Linear criteria applied to nonlinear aircraft.
   - The scope of MIL-F-8785C versus modern standards.
   - The cost of verification.

8. **Conclusions and future work.**

Expected contributions

C1. An aircraft-agnostic, frame-explicit model representation, verified to reproduce a hand-written reference model exactly.

C2. A gated, evidence-tagged verification method, including sabotage testing, demonstrated against public NASA check cases.

C3. A rule-based flying-qualities verdict with explicit margins, and a consistent definition of a robust pass under uncertainty.

C4. A demonstrated loop from verdict to design change on the F-16, with a path to distributed-propulsion aircraft.

References

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